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# Causal Structure Learning for Mitral Regurgitation via Graph Neural Networks

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**Abstract**

Mitral regurgitation (MR) arises from interdependent structural changes involving the mitral valve, left atrium, and left ventricle, yet most current machine learning approaches frame this inherently relational process as a single prediction task. This paper proposes a graph-based, causal framework for learning directed dependency structure around MR from echocardiographic measurements. We model eight clinically meaningful variables together with an explicit MR node in a masked directed graph, where the masking enforces admissible clinical directions and the final learned interaction matrix is visualized as an interpretable heatmap. Two architectures are studied: a one-layer masked graph model and a two-layer variant that reuses the same graph across two propagation steps to capture higher-order relations while preserving a single interpretable structure. On systolic echocardiographic data split at the patient level, the two-layer model shows lower error and yields a more structured clinically coherent influence graph. The learned heatmaps highlight contributions from atrial, annular, and ventricular measurements, suggesting that graph-based supervision can move MR analysis beyond scalar prediction toward directed structural interpretation.

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## 1. Introduction

Mitral regurgitation (MR) is among the most common clinically significant valvular disorders, and echocardiography remains the central noninvasive modality for evaluating valve anatomy, regurgitant severity, and the remodeling consequences on the left atrium (LA), mitral annulus, and left ventricle (LV) [1, 2, 3]. In current practice, MR assessment is strongly tied to geometric and Doppler-derived quantities such as chamber area, chamber length, chamber volume, tenting, annular size, vena contracta, and effective regurgitant orifice measures [1, 2]. At the same time, the pathophysiology of MR is fundamentally relational: annular dilation, leaflet tethering, atrial enlargement, and ventricular remodeling do not act in isolation, but through directed structural interactions that jointly shape the observed regurgitant burden [3, 4, 5].

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Recent artificial intelligence work on echocardiography has shown that machine learning can detect, stratify, and predict MR-related outcomes with promising accuracy [6, 7, 8, 9]. However, much of this literature is primarily predictive: the end goal is classification, survival prediction, or postoperative risk estimation. Such models are often useful clinically, but they usually provide limited insight into *how* echocardiographic variables interact as a directed system around the regurgitation process itself. For a problem such as MR, this limitation matters because clinicians are interested not only in whether MR is present or severe, but also in the anatomical and functional pathways that may explain it [3, 4, 10].

In parallel, graph neural networks (GNNs) and causal-learning methods have increasingly been combined to move from purely correlational modeling toward more structured and potentially more robust representations [11, 12, 13]. Graph-based learning is especially attractive in healthcare because medical variables are rarely independent; instead, they form systems of interacting measurements, structures, and processes [14, 15]. Recent surveys argue that causality-inspired GNNs can improve trustworthiness, interpretability, and out-of-distribution robustness by encouraging models to focus on mechanistically meaningful relations rather than spurious associations [11, 12, 13]. This motivates a graph-based, directed-structure perspective for MR.

The goal of this paper is therefore not to present another black-box MR predictor, but to formulate MR analysis as a *directed structure learning* problem centered on a clinically meaningful target. Starting from frame-level echocardiographic measurements, we learn a masked directed interaction matrix whose last row quantifies how the input features influence MR area. The learned graph is then visualized as a heatmap, yielding an interpretable output object rather than only a scalar prediction. In this way, prediction is used as a supervisory signal for discovering directed dependency patterns.

Two closely related approaches are investigated. The first is a single-layer masked graph model that learns a direct feature-to-feature and feature-to-MR propagation operator. The second reuses the same masked graph across two propagation steps separated by a nonlinear activation, allowing the model to capture higher-order interactions while preserving a single interpretable graph. While the first approach mainly aims to capture direct, linear relationships, the second approach is expected to better capture indirect pathways, which is important in MR where anatomical changes often influence regurgitation through intermediate remodeling effects [4, 5].

The resulting paper is positioned at the intersection of echocardiographic MR analysis, graph learning, and causality-inspired modeling. The learned directed graph should be interpreted as a *causal-motivated dependency structure*, not as a definitive interventionally validated causal graph. Unlike classical causal discovery methods that enforce full acyclicity or rely on stronger structural assumptions [16, 17], the present framework is supervised by MR prediction and feature reconstructability, and constrained only by a clinically designed mask. Its contribution is thus a practically interpretable, graph-based MR modeling pipeline that moves beyond pure prediction and toward directed structural explanation.

## 2. Related Work

Graph-based deep learning has become an important direction in medical AI because it can represent anatomical, physiological, and clinical entities as interacting nodes rather than isolated covariates. Reviews of graph learning in medicine highlight its usefulness for modeling non-Euclidean structure, multimodal relations, and clinically meaningful dependencies that standard tabular or convolutional models may fail to capture [14, 15]. In particular, healthcare applications increasingly use GNNs for risk prediction, patient similarity modeling, and representation learning from irregular clinical data [14, 15]. This broader trend supports the idea that cardiac measurements should also be modeled relationally.

A second relevant line of work concerns the integration of causality with graph learning. Classical causal inference provided the conceptual language of directed relations, interventions, and structural assumptions [16]. More recently, several surveys have synthesized the emerging literature on causality-inspired GNNs, emphasizing their potential to improve robustness, explainability, and transferability [11, 12, 13]. The recent review of causal GNNs in healthcare explicitly argues that the combination of graph structure and causal reasoning is particularly relevant in clinical settings where spurious correlation, confounding, and distribution shift are major concerns [13]. Similarly, benchmarking work on causal integration in GNNs reports that causal design choices can improve model robustness and downstream classification performance [18].

Despite this progress, the medical GNN literature and the causal-GNN literature only partially overlap with the specific needs of echocardiographic MR analysis. Much of the healthcare graph literature focuses on electronic health records, molecular graphs, or imaging cohorts with patient-level graph constructions [14, 15]. By contrast, our setting is a compact physiological variable graph in which each node corresponds to a clinically meaningful echocardiographic measurement, and the main output is not a label alone but a directed weighted interaction structure centered on MR.

MR-specific AI work provides an important but mostly predictive baseline. Kang et al. used machine learning to predict survival and recurrent MR after mitral valve repair, showing that nonlinear models can outperform conventional survival modeling in postoperative outcome prediction [6]. Long et al. proposed DELINEATE-MR, a deep learning system for grading MR severity directly from full transthoracic echocardiography studies [7]. Vrudhula et al. developed a large-scale automated deep learning pipeline for detecting clinically significant MR from Doppler echocardiographic videos [8]. Sadeghpour et al. presented an automated multiparametric workflow for MR severity grading that combines several echocardiographic variables into a high-performing machine learning tool [9]. These studies demonstrate that MR is amenable to machine learning and that automated quantification is clinically promising. However, their primary outputs remain risk scores, severity labels, or outcome predictions rather than an interpretable directed graph over the constituent echocardiographic variables.

From the clinical side, the mechanistic literature emphasizes that MR results from structured interactions among the mitral annulus, leaflets, chordae, papillary muscles, atrium, and ventricle [3]. In secondary or functional MR, atrial and ventricular remodeling are especially relevant: LA enlargement can alter annular geometry, while LV remodeling can displace papillary muscles and increase leaflet tethering, both reducing coaptation and increasing regurgitation [4, 5, 10]. This makes MR particularly suitable for a directed graph formulation. Instead of treating the measurements merely as independent predictors, one can attempt to learn how they organize into a weighted influence structure around the target regurgitant area.

The present work is positioned relative to these strands as follows. Compared with predictive MR studies, it shifts the focus from "predicting MR" to "learning a directed MR-centered structure". Compared with general causal discovery, it does not claim full causal identification, but instead proposes a causality-inspired supervised structure learner tailored to a clinically constrained set of echocardiographic variables. Compared with broader healthcare GNN work, it uses an intentionally small and interpretable node set whose semantics are directly meaningful to cardiologists.

## 3. Proposed Approach

### 3.1. Problem setting

Let

$$\mathbf{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d \tag{1}$$

denote the vector of echocardiographic input features for one frame, where in our case $d = 8$ and the variables are left atrium (LA) area, length, and volume, mitral valve (MV) tenting height and annulus, left ventricle (LV) area, length, and volume. The supervised target is the MR area

$$y \in \mathbb{R}_{\geq 0}, \tag{2}$$

measured in $\mathrm{cm}^2$.

Rather than learning a direct map $\mathbf{x} \mapsto y$ only, we augment the system with an explicit MR node and learn a directed weighted interaction matrix over all $d+1$ nodes. The final node corresponds to MR, while the first $d$ nodes correspond to the input features:

$$\mathbf{h}^{(0)} = [x_1, \ldots, x_d, 0] \in \mathbb{R}^{d+1}. \tag{3}$$

The zero initialization of the MR node means that MR is not fed back into the graph as an input feature. Instead, it is generated from the other nodes through learned directed propagation.

This design is motivated by a causal viewpoint. In a clinically meaningful directional interpretation, geometric and chamber variables may influence MR, while the reverse direction is not used in the model. Accordingly, the

architecture is constrained so that feature-to-feature edges are allowed, feature-to-MR edges are allowed, self-loops are removed, and MR-to-feature edges are forbidden. The result is a directed, target-centered graph whose last row directly quantifies the learned incoming influence on MR.

### 3.2. Entry data and preprocessing

The raw spreadsheet contains 1003 frame-level observations from 26 patients and multiple recorded measurements per frame. Following the proposed pipeline, only systolic frames were retained, yielding 831 observations. This is clinically consistent with MR being a systolic regurgitant phenomenon and with systolic geometric configuration being directly relevant to the measured jet area [1, 3]. The retained variables are summarized in Table 1.

Table 1. Data summary used by the proposed pipeline.

| Quantity | Value |
|---|---|
| Total raw observations | 1003 |
| Systolic (filtered) observations | 831 |
| Unique patients | 26 |
| Input features | 8 |
| Target | MR area ($cm^2$) |
| Missing values after filtering | 13 values for MV tenting height |
| Train / validation / test split (patient level) | 64% / 16% / 20% |

To avoid leakage across repeated measurements from the same patient, the split was performed at the patient level. First, an outer 80/20 split separated train-validation data from test data. Then, the train-validation subset was split again with the same 80/20 grouped strategy into training and validation partitions. This choice is essential because frame-level samples from the same patient are highly dependent; a naive random split would overestimate generalization performance.

Preprocessing was intentionally simple and fully reproducible. Missing values in the feature matrix were imputed with a median imputer, which is appropriate here because missingness was limited and concentrated in one variable. The input features were then standardized to have zero mean and unit variance. The target MR area is *not* standardized in our proposed pipeline; hence, all reported target losses remain in raw squared target units. This is especially useful for interpreting the numerical results in Section 4. The preprocessing sequence is consistent with common echocardiography machine learning practice, where normalization is applied to heterogeneous geometric measurements while preserving direct clinical interpretability of the final target.

### 3.3. Graph parameterization and masking

Let $\mathbf{W} \in \mathbb{R}^{(d+1)\times(d+1)}$ denote the unconstrained learnable weight matrix and let $\mathbf{M} \in \{0,1\}^{(d+1)\times(d+1)}$ be a fixed binary mask encoding the clinically admissible directions. The effective graph is

$$\mathbf{A} = \mathbf{W} \odot \mathbf{M}, \tag{4}$$

where $\odot$ denotes elementwise multiplication.

The mask is defined as

$$M_{ij} = \begin{cases} 1, \; i \le d, \; j \le d, \; i \ne j, \\ 1, \; i = d+1, \; j \le d, \\ 0, \; \text{otherwise.} \end{cases} \tag{5}$$

Under the row-target / column-source convention used in the heatmaps, row $i$ collects the incoming contributions into node $i$ from source nodes $j$. Therefore, the last row corresponds to the directed influence of the eight echocardiographic features on MR.

This masking plays two roles. First, it injects domain knowledge into the model by excluding clinically implausible directions under the intended interpretation. Second, it makes the learned graph easier to interpret than a dense unconstrained matrix. Using the masking, the architecture can therefore be viewed as a supervised causal structure learner.

### 3.4. Approach 1: single-layer masked graph model

The first approach applies one graph propagation step to the augmented node vector:

$$\mathbf{h}^{(1)} = \mathbf{h}^{(0)}\mathbf{A}^\top + \mathbf{b}, \tag{6}$$

where $\mathbf{b} \in \mathbb{R}^{d+1}$ is a learnable bias vector.

The MR prediction is extracted from the last component and passed through a softplus transformation:

$$\hat{y} = \text{softplus}\left(h^{(1)}_{d+1}\right). \tag{7}$$

The softplus ensures a nonnegative output, which is appropriate because MR area is nonnegative by definition. The first $d$ components of $\mathbf{h}^{(1)}$ are retained as reconstructed feature outputs:

$$\mathbf{z} = \left(h^{(1)}_1, \ldots, h^{(1)}_d\right). \tag{8}$$

Training combines a target regression objective with a structural reconstruction objective:

$$\mathcal{L}_{\text{task}} = \frac{1}{N}\sum_{n=1}^{N}(\hat{y}_n - y_n)^2, \tag{9}$$

$$\mathcal{L}_{\text{struct}} = \frac{1}{N}\sum_{n=1}^{N}\|\mathbf{z}_n - \mathbf{x}_n\|_2^2, \tag{10}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{struct}}. \tag{11}$$

The auxiliary reconstruction term ($\lambda$) regularizes the graph so that it must also preserve the internal feature structure, rather than only fitting MR. In practical terms, it discourages degenerate solutions in which the model ignores clinically meaningful feature interactions and focuses only on a narrow subset of edges. We set $\lambda = 20$ empirically, as this value yielded a good trade-off between preserving inter-feature structure and maintaining strong performance on the MR prediction objective.

The single-layer model is the simplest possible directed graph learner in the present setting. Each target node receives one weighted mixture of the source nodes, and MR is predicted directly after one propagation step. This makes the learned graph highly transparent, but it also limits the model to first-order interactions.

### 3.5. Approach 2: two-layer masked graph model

The second approach preserves the same masked graph matrix $\mathbf{A}$ but applies it twice, with a nonlinear activation between the two propagations. Specifically,

$$\mathbf{h}^{(1)} = \sigma\left(\mathbf{h}^{(0)}\mathbf{A}^\top + \mathbf{b}^{(1)}\right), \tag{12}$$

$$\mathbf{h}^{(2)} = \mathbf{h}^{(1)}\mathbf{A}^\top + \mathbf{b}^{(2)}, \tag{13}$$

where $\sigma(\cdot)$ is a LeakyReLU activation and $\mathbf{b}^{(1)}, \mathbf{b}^{(2)} \in \mathbb{R}^{d+1}$ are learnable bias vectors. The MR output is then

$$\hat{y} = \text{softplus}\left(h^{(2)}_{d+1}\right), \tag{14}$$

and the reconstructed features are given by the first $d$ components of $\mathbf{h}^{(2)}$.

The key design choice is that *the same graph matrix* is reused in both propagation stages. This is beneficial for interpretability: the model still produces one directed graph rather than two unrelated matrices. At the same time, the second propagation allows information to travel along longer paths. Therefore, the two-layer model can represent effects such as

$$x_j \to x_k \to y, \tag{15}$$

which are inaccessible to the one-layer version unless they can be collapsed into a single direct edge. In physiological terms, this matters because MR may arise through intermediate remodeling pathways, not only through direct one-step associations. For instance, chamber enlargement may alter annular geometry, which in turn affects regurgitation [4, 5, 10].

The total loss remains the same as in Eqs. (9)–(11), so the comparison between approaches isolates the effect of adding one nonlinear propagation step while keeping the same target, preprocessing, masking principle, and structural regularization. Conceptually, this makes the comparison especially clean. If the second model performs better, the improvement can be attributed to its ability to encode higher-order directed interactions while retaining graph-level interpretability.

### 3.6. Training protocol and output type

Both models were trained with Adam using a learning rate of $5 \times 10^{-4}$. The maximum number of epochs was 2000, with early stopping based on validation task loss using patience 50 and minimum improvement threshold $10^{-4}$. The selection criterion was the best validation loss on MR prediction, after which the best model state was restored and evaluated on the held-out test set.

The final output of each approach is *not* just the scalar MR prediction. Instead, the main explanatory output is the learned directed graph

$$\mathbf{A} = \mathbf{W} \odot \mathbf{M}, \tag{16}$$

which is converted to a heatmap after taking absolute values and row-wise normalization:

$$\widetilde{A}_{ij} = \frac{|A_{ij}|}{\sum_{k=1}^{d+1} |A_{ik}| + \varepsilon}, \tag{17}$$

with a small $\varepsilon > 0$ for numerical stability.

Thus, in the heatmap matrix, rows correspond to target nodes, columns correspond to source nodes, and the final row represents how the eight features influence MR. As a result, each trained model yields an interpretable weighted directed graph that can be examined clinically. This graph-centered output is what makes the pipeline causality-inspired: the model is supervised by MR, but the main scientific object of interest is the learned dependency structure.

It is worth emphasizing that the present method is not a full DAG-based causal discovery algorithm in the sense of Pearl or NO TEARS [16, 17]. No global acyclicity constraint is imposed among the feature nodes, and the model is optimized under a supervised objective tied to MR regression. The learned graph should therefore be interpreted as a clinically grounded causal-motivated graph.

## 4. Numerical Experiments

### 4.1. Experimental objective

The experiments address two questions. First, can the proposed graph-based models achieve meaningful predictive performance for MR area from the selected echocardiographic variables? Second, and more importantly for this paper, do the resulting directed graphs reveal clinically plausible influence patterns around MR? Because both approaches share the same preprocessing, data split, target, and regularized objective, the comparison isolates the value of adding a second nonlinear propagation step.

### 4.2. Target-loss comparison

Table 2 reports representative validation and test losses obtained across the notebook runs.

Across repeated runs, the two-layer model generally shows better validation and test performance. Relative to the one-layer variant, it yields noticeably lower validation and test error while preserving interpretability: the second model still produces a clinically readable directed graph. This consistent improvement supports the hypothesis from Section 3 that a second propagation step helps the model represent higher-order relations among atrial, annular, and ventricular measurements before mapping them to MR.

Table 2. Comparison of the two approaches on the MR target. The reported metric is MSE on raw MR area; since MR area is measured in $cm^2$, the MSE is in squared target units. RMSE is included for interpretability. The values shown are representative of typical runs.

| Approach Validation MSE | Test MSE | Test RMSE | |
|---|---|---|---|
| Approach 1: one-layer masked graph | 21.39663 | 3.44691 | 1.8566 |
| Approach 2: two-layer masked graph | 15.08099 | 2.77839 | 1.6669 |

### 4.3. Heatmap outputs

Figures 1 and 2 visualize the directed influence matrices learned by the two proposed approaches.

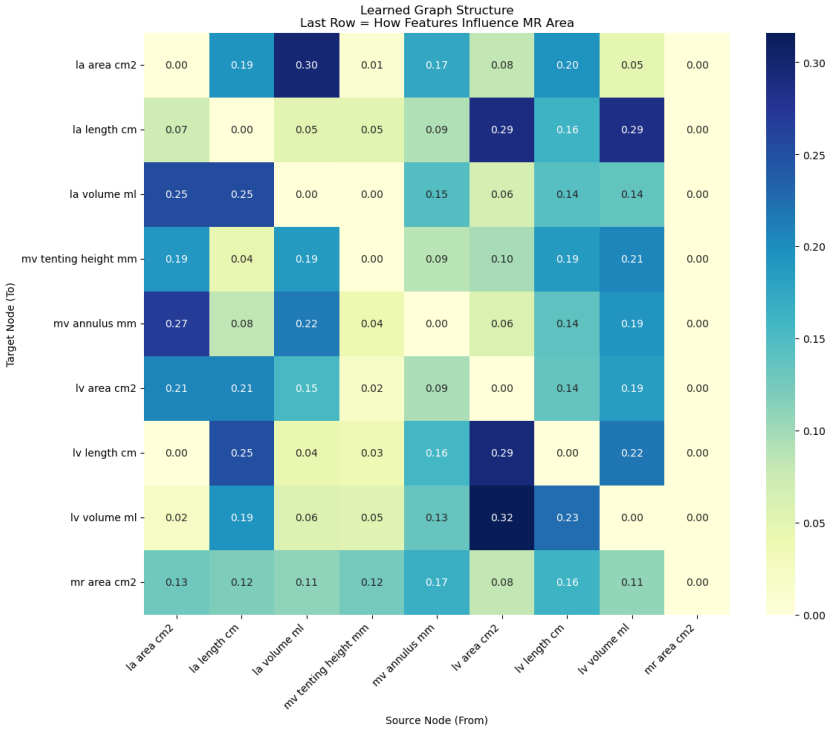


Fig. 1. Heatmap produced by Approach 1 (single-layer masked graph model). Rows correspond to target nodes and columns correspond to source nodes. The last row represents the learned influence of the eight echocardiographic features on MR area.

### 4.4. Interpretation of Approach 1

The one-layer heatmap already reveals a meaningful directed structure rather than a trivial or collapsed solution. In the MR row, the influence is distributed across several variables, with visibly larger weights assigned to MV annulus, LV length, and LA area. This matches clinical expectations at a broad level. Annular geometry, leaflet tethering, and chamber remodeling are all known to be involved in functional MR mechanisms [3, 4, 5]. The fact that no single feature dominates completely is also reasonable, since MR severity is rarely governed by one isolated measurement.

At the same time, the first approach is somewhat diffuse. The MR row spreads its influence mass relatively evenly across many features, which makes the graph interpretable but less selective. This is consistent with the single-layer design. Because MR is predicted after one linear masked propagation, the model can only express first-order dependencies. As a result, different mechanisms may be blended together into a flatter influence pattern.

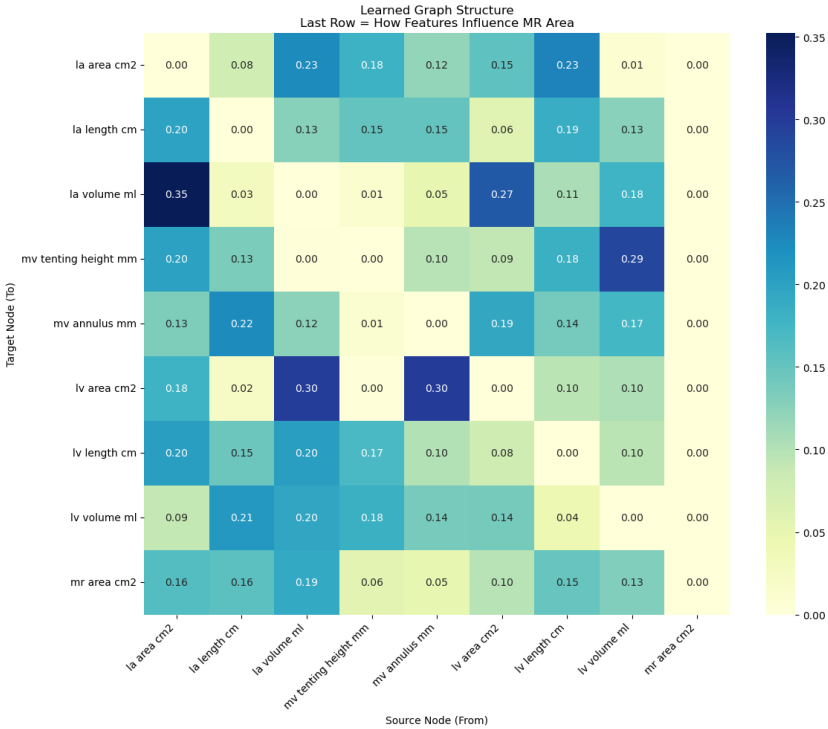


Fig. 2. Heatmap produced by Approach 2 (two-layer masked graph model). The output interpretation is identical to Fig. 1.

### 4.5. Interpretation of Approach 2

The two-layer model produces a more structured graph while also improving target loss. In the MR row, the strongest weights shift toward LA volume, LA area, LA length, and LV length, with additional contributions from LV volume and LV area. By contrast, MV tenting height and MV annulus receive smaller final-row weights than in the one-layer case. This can be interpreted in two complementary ways.

First, the second model appears to attribute more of the final MR signal to chamber-remodeling variables, especially the LA and LV size descriptors. Clinically, this is plausible because atrial and ventricular remodeling are central drivers of functional MR, influencing annular shape, coaptation, and tethering through intermediate pathways [4, 5, 10]. Second, because the two-layer model can propagate information through intermediate feature nodes, it may be representing annular and tethering effects *indirectly* through broader remodeling pathways rather than forcing them to appear as dominant direct one-step inputs into MR.

This is exactly where the two-layer design becomes attractive. It does not merely fit the target better; it also produces a graph that can encode clinically coherent indirect relations. For example, LA and LV enlargement can be viewed as higher-level remodeling descriptors whose interaction with annular and leaflet geometry unfolds over multiple mechanistic steps. A two-step propagation is better aligned with such a picture than a one-step linear graph. Therefore, the better loss of Approach 2 and the more structured appearance of its heatmap reinforce one another.

### 4.6. Clinical consistency of the learned influences

The learned influences should be interpreted against established MR pathophysiology. Several broad observations are encouraging.

First, both models assign nonzero importance to LA-related, annular, and LV-related variables, which is expected. The literature emphasizes that MR is linked to the geometry of the entire mitral apparatus and the surrounding chambers, not to a single isolated scalar [3, 4]. The presence of the nonzero weights is also consistent with the fact that

LA and LV measurements are strongly coupled through cardiac remodeling, and annular behavior is not independent of chamber configuration [3, 5]. Second, the two-layer graph's stronger emphasis on LA and LV size variables is compatible with the view that regurgitation burden reflects the broader structural remodeling of the left heart rather than only local valve geometry [4, 10].

Naturally, these findings should be interpreted with appropriate caution. The model is trained on a relatively small number of patients, and the graph is supervised by prediction rather than identified from interventional data. Consequently, the heatmaps are best interpreted as learned directed dependency patterns, which can then be compared with established clinical knowledge of MR.

## 5. Conclusions

This paper formulates MR analysis as a directed graph-learning problem rather than as pure outcome prediction. Starting from frame-level echocardiographic measurements, we construct a clinically constrained graph whose nodes are interpretable variables and whose final row quantifies learned influences on MR area. This produces an explanatory output object (a directed heatmap) that complements the scalar regression task.

Two models were studied. The first learns a one-step masked propagation from echocardiographic variables to MR, while the second adds a second nonlinear propagation step using the same graph. The two-layer model achieves lower validation and test losses and yields a more structured, clinically coherent influence pattern. In particular, it better captures the idea that MR emerges from interacting chamber-remodeling and valve-geometry pathways rather than from isolated direct effects.

The main contribution of the work is therefore methodological and interpretive. The proposed pipeline offers a simple but effective bridge between predictive MR modeling and causal structure reasoning. It does not yet establish a fully identified causal graph in the strict interventional sense, but it provides a directed, clinically grounded graph that can support explanation and hypothesis generation.

Several extensions follow naturally. Future work could incorporate explicit acyclicity constraints, as well as temporal ordering across the cardiac cycle. Even in its current form, however, the approach shows that a graph-centered perspective can reveal interpretable structure around MR and that an increase in architectural depth can materially improve both predictive and explanatory quality.

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